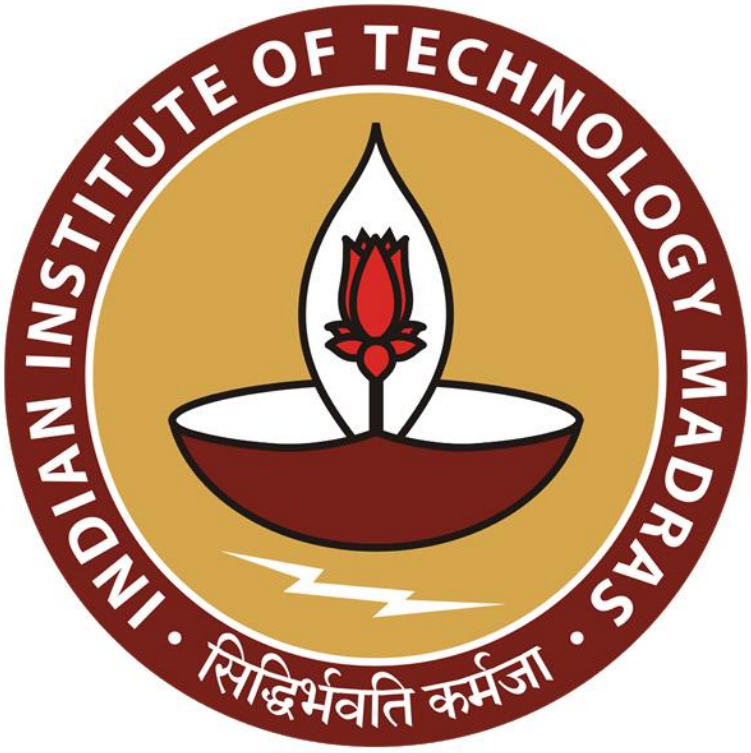


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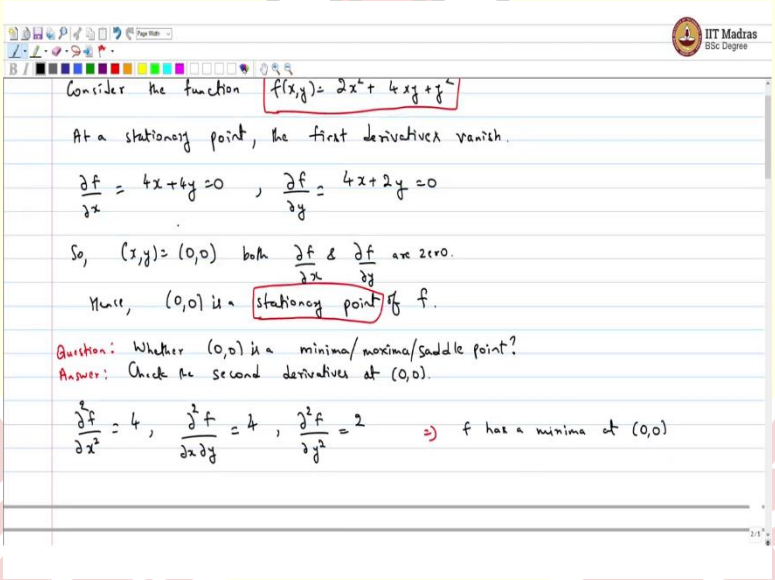
ONLINE DEGREE

Machine Learning Foundations
Professor Prashanth L A
Department of Computer Science & Engineering
Indian Institute of Technology, Madras
Positive Definiteness

Welcome to week 6 of Machine Learning Foundations. In this week, we will study two important concepts in linear algebra. The first is positive definiteness. We will look at it in the context of functions and then matrices, and then we will move on to the most important topic, which concludes this linear algebra segment, which is principal component analysis.

In the first lecture today, we will understand what positive definiteness is, in the context of functions. We will define it and then make a connection to linear algebra by looking at vectors and suitable matrices. In the following lecture, we will extend this notion to matrices to understand what it means, what one means to know when he says, when he or she says, a matrix is positive definite.

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We are looking at Positive Definite Matrices. But before that, I will motivate this using very simple quadratic functions. So, consider the function f (x , y) = $2x^{2.}$ +4xy+y^2 So, we want to understand what happens to this function when I differentiate.

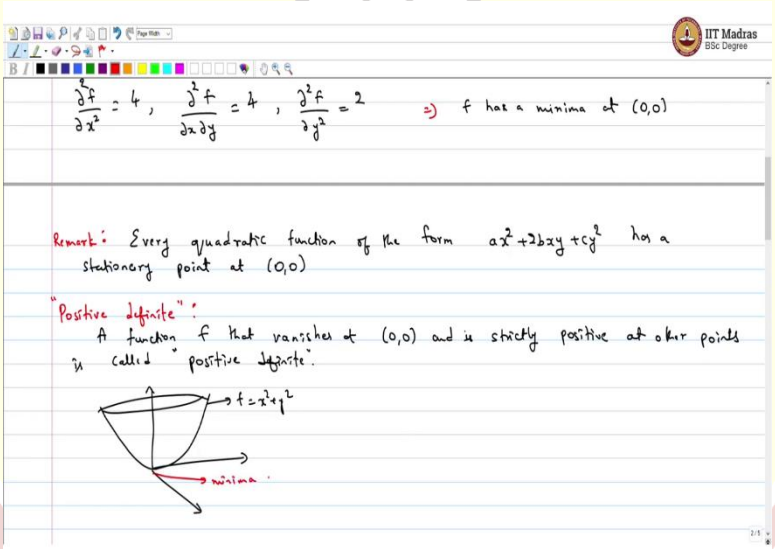
So, recall, at a stationary point of any function the first derivatives vanish. In this case, it is a function of two variables, so both the partial derivatives have to vanish. So, if I do $\frac{\partial f}{\partial x}$ then that is 4x + 4y and that has to vanish, and do $\frac{\partial f}{\partial y}$ is 4x + 2y and that has to vanish. Now, it is

easy to see that. So, if I set (x, y) to be the origin, which is (0, 0) both do $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ are 0 hence (0, 0) is a stationary point of f.

Now, the question is, so what we have just found a point where the first derivative vanishes. So, if you recall from high school, it is not clear, the important question now is, , whether (0, 0) is a minima or maxima or saddle point?

Now, for this, what you have to do is, the answer to this question, answer, check the second derivatives, check the second derivative at (0, 0). So, all you have to do is $\frac{\partial^2 f}{\partial x^2}$ and that is very simple. You just have to differentiate this with respect to x again. So, and similarly, I mean, so this would turn out to be 4. Now, $\frac{\partial^2 f}{\partial x\, \partial y}$ turns out to be 4, and $\frac{\partial^2 f}{\partial y^2}$ turns out to be 2.

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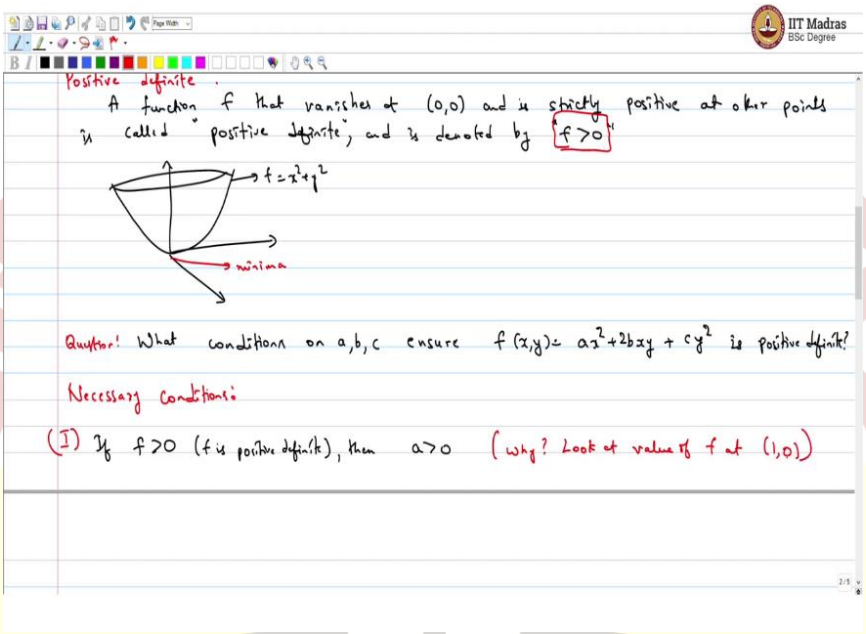
Now, we will try to understand. So, the before delving deeper, let me just make a quick remark. I mean, this answer, implies f has minima at (0, 0). We will argue this more rigorously, very soon, but let me just write a very general remark, every quadratic function of the form

$ax^2 + 2bxy + cy^2$ has a stationary point.

If you are not claiming that it is minimized has a stationary point, if you just differentiate the (4, 4) into coefficients do not matter when I take the derivative then check the partial or the first derivatives at (0, 0), irrespective of the coefficient, irrespective of the coefficient in $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$, whether it is 4x + 4y or some other coefficient it does not matter. If I put x = y = 0, then I see that the derivative vanishes.

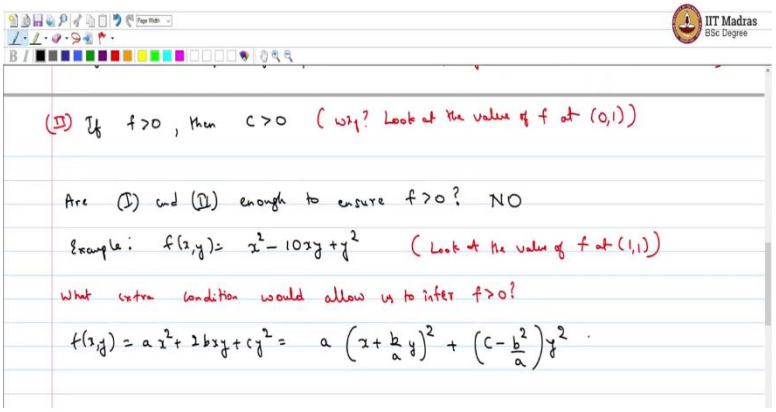
So, it is clear that for every quadratic function has a stationary point at (0, 0). Now, to the definition of positive definite, so a function f that vanishes at (0, 0) and is strictly positive at other points is called positive definite. Now, if you want a rough geometric figure, so what we have is a function like this. This is $f = x^2 + y^2$. And at origin this is a minima.

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So, the question that we are going to ask is, question, what conditions on a, b, c ensure $f(x, y) = ax^2 + 2bxy + cy^2$ is positive definite. I mean, obviously, we will look at some necessary conditions, some obvious necessary conditions. These are not sufficient to understand if the very first condition is, yeah, there is some dou, and is denoted by $f > 0$. So, that this shows that $f > 0$, which is like saying, f is positive definite, then $a > 0$ (a has to be greater than 0). Now, I would like you to think about this and look at value of f at (1,0). Now 1, 0 is the value of f, it is just a. And since it is positive definite it has to be > 0.

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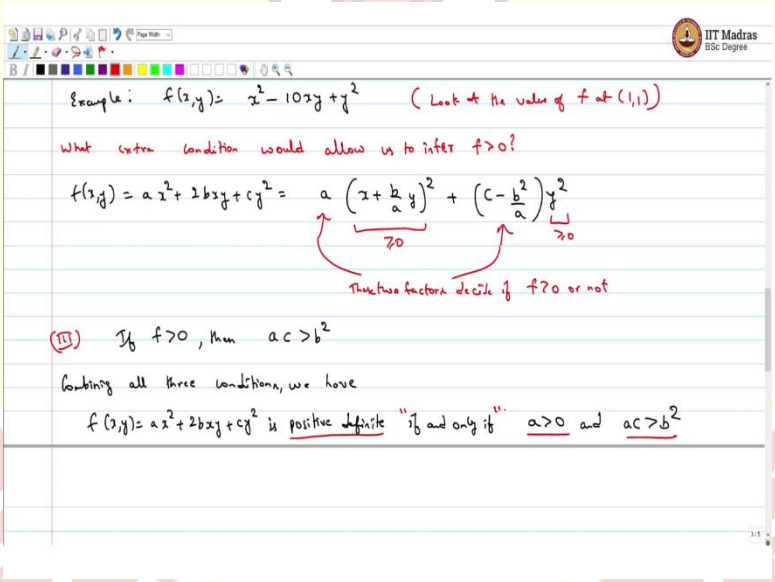
Now, along same logic if f > 0 then c > 0. I mean, again, it is same. Why? Look at the value of f at (0, 1). Now, are 1 and 2 enough to ensure f > 0?

Again, pause and think about this. Think, if I have a function $ax^2 + 2bxy + cy^2$ with both a and c > 0 is it already enough to infer that f is positive definite? The answer is, no. If you want an example, $f(x, y) = x^2 - 10xy + y^2$. So here is a function which is not positive definite, it is definitely not positive definite.

You can look at, I mean, you can easily make this function go below 0. I mean, again, the value of f at (1, 1) it is clearly less than 0. What extra condition would allow us to infer f > 0? For this, what you have to do is, counted the squares. So, what you can do is, you can write

$f(x,y)= ax^2 + 2bxy + cy^2 = a(x + \frac{b}{a}y)^2, + (c - \frac{b2}{a}) \times y^2$.

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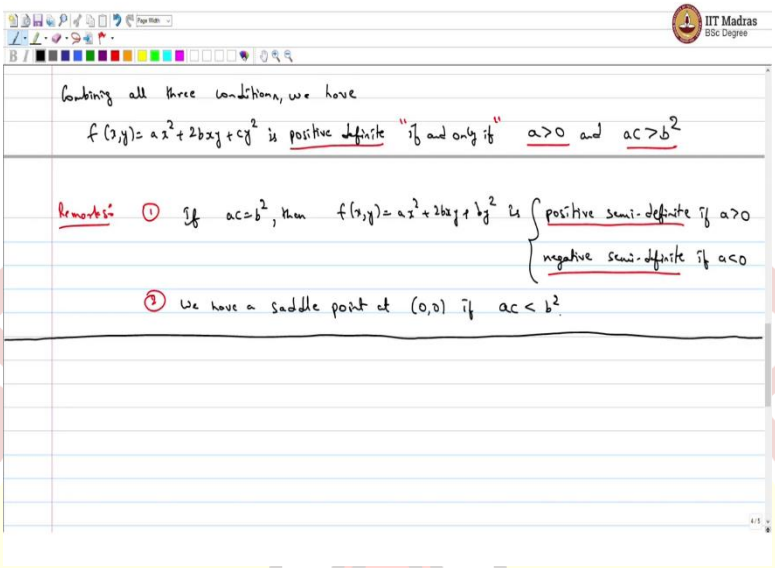


So, that $\frac{b^2}{a} \times y^2$ root get canceled and we will recover whatever was the original function. Now, it is easy to see that, $\frac{x + b}{a}$ so this is a positive term. And this is also a positive term. So, it comes down to what is the value of a and what is the value of $(x + \frac{b}{a}y)^2$ is ≥ 0, $(c - \frac{b2}{a}) \geq 0$, so these two factors, decide if f > 0 or not. So, that brings us to the third condition.

So, the third condition is, if f > 0, then it is clear that a has to be greater than 0, and that is something we already saw. So, what $(c - \frac{b2}{a})$ here has to be greater than 0, which is like saying

that $ac > b^2$. So, combining all three conditions, we have $f(x, y) = ax^2 + 2bxy + cy^2$ is positive definite if and only if $a > 0$ and $ac > b^2$.

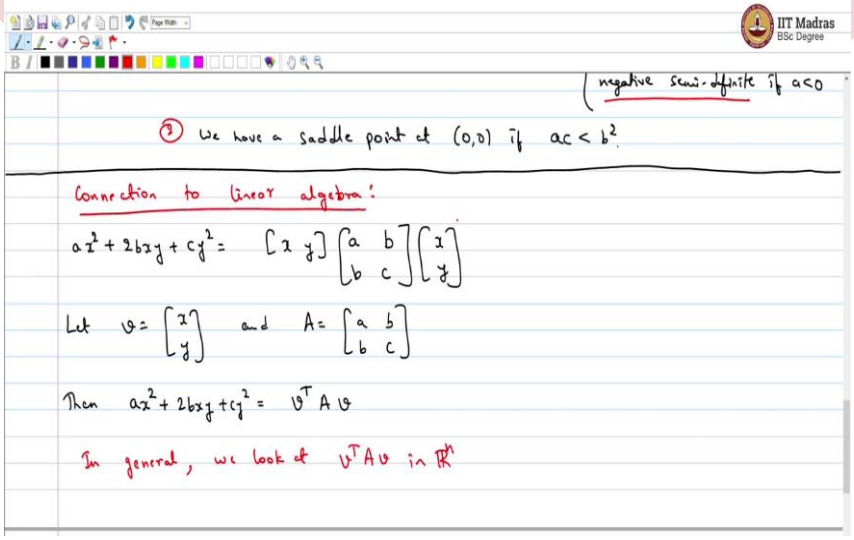
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So if it is positive definite, it is clear that the origin it is minimized, that is clear. Just a couple of remarks. If $ac = b^2$ then f of this form, which is f is again this quadratic form $f(x, y) = ax^2 + 2bxy + cy^2$ is positive semi-definite. We call it semi-definite if $a > 0$, and negative semi-definite if a is less than 0. The second point is we have a saddle point at (0, 0) if ac is less than b^2.

Now, this is, so far, we do not have any linear algebra really in this. Just we are trying to understand whether origin is a point where the function is minimized for a particular class of functions, which is a quadratic function. And we saw that if $a > 0$, and if $ac > b^2$, then 0, 0 is not just a stationary point, but it is also a point where the function is minimized.

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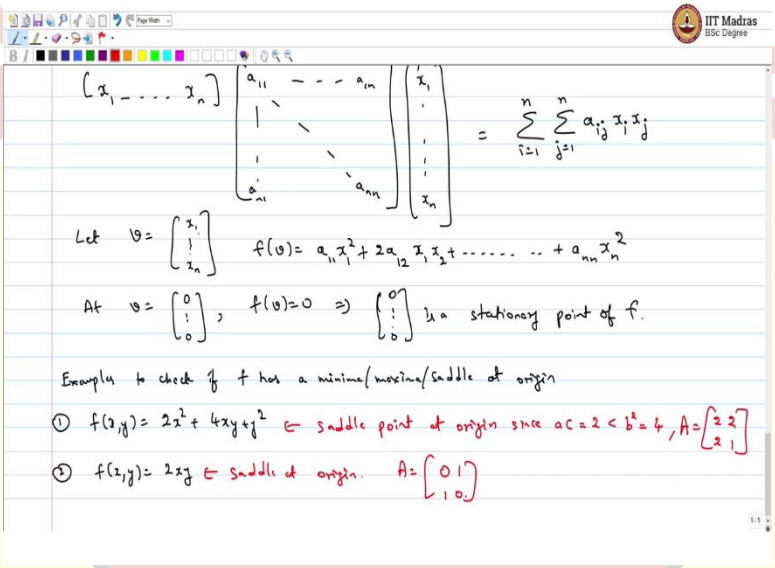


Now, we will see the most important connection to linear algebra. Here is where we will bring in matrices and dot products and so on. So, I will illustrate this in again the quadratic case and then generalize. Now, $(ax^2 + 2bxy + by^2)$this can be written as $[x, y]\begin{bmatrix} a & b \\ b & c \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$.

Now, I will denote column vector by V = $\begin{bmatrix} x \\ y \end{bmatrix}$, and let A denote matrix $\begin{bmatrix} a & b \\ b & c \end{bmatrix}$, then

$ax^2 + 2bxy + cy^2 = V^T AV$. And the question about positive definiteness of $ax^2 + 2bxy + cy^2$ we will try to relate it to this matrix A very soon, but now in general, we look at $V^T AV$ in $\mathbb{R}^{th}$

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So, the way it would be is, if you look at this vector [x1…… to xn] $\times \begin{bmatrix} a_{11} & \cdots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{nn} \end{bmatrix} \times$[x1…… to xn]^T. Then I can write this as $\sum_{\substack{1 \le i \le n \\ 1 \le j \le n}} a(i,j)x_i x_j$.

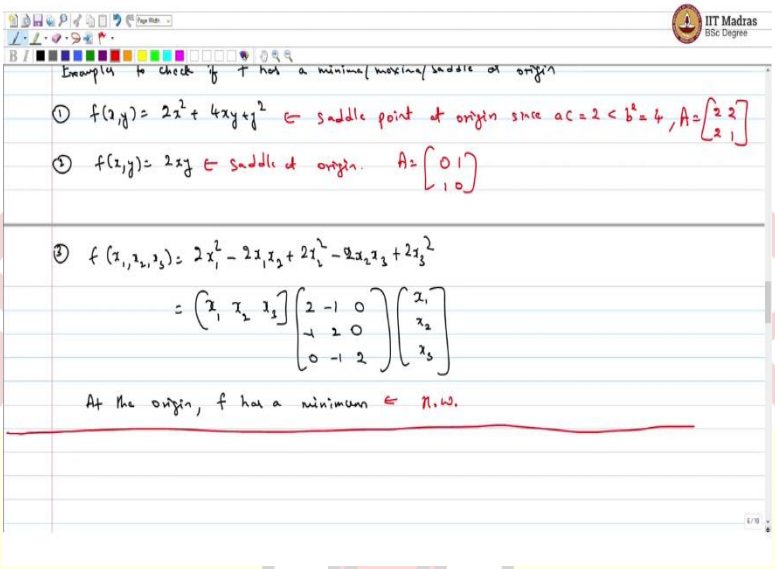
Now, let V be [x1…… to xn]^T and $f(V) = a_{11} x_1^2 + 2a_{12} x_1 x_2 + \ldots$..so on….+ $a_{nn} x_n^2$. Now, it is clear that at V= $\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$ f (v)= 0 implies is a $\begin{bmatrix} 0 \\ \vdots \\ 0 \end{bmatrix}$ is a stationary point of f.

Few examples check if f has a minima/ maxima / saddle at the origin.

First example, imagine f(x, y) just two variables $2x^2 + 4xy + y^2$. Now in this case, you can infer as a saddle point at origin. Why? Since ac =2 is less than b^2=4. The second example is a much simpler example f(x, y) = 2xy. I mean, for this case if I write this in the dot product then the

matrix A turns out to be $\begin{bmatrix} 2 & 2 \\ 2 & 1 \end{bmatrix}$. This also has a saddle at origin for the same condition and the A matrix that makes up $V^T AV$ is $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$.

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In final example before I define positive definiteness in the context of a matrix, and this is imagine $f(x_1, x_2, x_3) = 2x_1^2 - 2x_1x_2 + 2x_2^2 - 2x_2x_3 + 2x_3^2$. This can be written as $[x_1, x_2, x_3]$ into some matrix $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$. And that matrix turns out to be $\begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & 0 \\ 0 & -1 & 2 \end{bmatrix}$. Now, at the origin f has a minimum, and this is a homework, check this.

A recap before we end this lecture. We started looking at the notion of positive definiteness in the context of functions. And in taking a simple quadratic function we understood when this function is definitely positive, and what are the necessary conditions. And we also figured, what are conditions that are both necessary and sufficient for a function to be positive definite.

We made a connection to linear algebra by using matrices and vectors to connect this notion of positive definiteness using matrices, that is the linear algebra viewpoint. In the next lecture, we will use this connection to come up with a natural definition of positive definiteness in the context of matrices. We will define when a real matrix is positive definite and use this notion. I mean, we will also try to understand some properties of positive definite matrices in the next lecture.